

Homework 4

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Solution 4.1.

1. $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, AB \neq BA.$
2. $(A+B)^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, A^2 + 2AB + B^2 = \begin{bmatrix} 6 & 4 \\ 4 & 4 \end{bmatrix}, (A+B)^2 \neq A^2 + 2AB + B^2.$
3. $(AB)^2 = ABAB = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}, A^2B^2 = AAB B = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}, (AB)^2 \neq A^2B^2.$

Solution 4.2.

1. $2A - I, \quad 4A + 4I.$

2. Since the inverse is also a polynomial of A , and from 1, all polynomial of A can be simplified to $sA + tI$, set $(I + 2A)^{-1} = xI + yA$. By the definition of the inverse,

$$(I + 2A)(xI + yA) = I \implies xA + yA + 2xI + 2yA = I \implies \begin{cases} 3y + 2x = 0 \\ x = 1 \end{cases} \implies x = 1, y = -\frac{2}{3}$$

So $(I + 2A)^{-1} = I - \frac{2}{3}A.$

Solution 4.3.

1. The multiplication is equivalent to one swap of 1, 3 columns and one swap of 1, 3 row, so $\begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}.$
2. The multiplication is equivalent to using the first row to kill the second and the third row, so $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 2 & -3 \end{bmatrix}.$
3. The multiplication is equivalent to using the first row to kill all rows below, so $\begin{bmatrix} 1 & & & & \\ 0 & 1 & & & \\ 0 & 2 & 1 & & \\ 0 & 3 & 3 & 1 & \\ 0 & 4 & 6 & 4 & 1 \end{bmatrix}.$
4. if $k = 4n, I_4.$ if $k = 4n + 1,$ $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$

$$\text{if } k = 4n + 2, \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \quad \text{if } k = 4n + 3, \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \text{ where } n = 0, 1, 2, \dots$$

Solution 4.4.

$$1. X_{13} = e_1 e_3^T = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad X_{32} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad X_{12} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$2. X_{13}X_{32} = e_1 e_3^T e_3 e_2^T = e_1 e_2^T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$X_{32}X_{13} = e_3 e_2^T e_1 e_3^T = O.$$

$$3. \text{ if } i \neq j, \text{ then } X_{ij} = e_i e_j^T e_i e_j^T = O.$$

4.

Proof:

First, we prove that $AB = BA \implies (A - I)(B - I) = (B - I)(A - I)$.

If $AB = BA$, then

$$\begin{aligned}(A - I)(B - I) &= AB - A - B + I \\ &= BA - A - B + I \\ &= (B - I)(A - I)\end{aligned}$$

Next, we prove that $(A - I)(B - I) = (B - I)(A - I) \implies AB = BA$.

$$\begin{aligned}(A - I)(B - I) &= (B - I)(A - I) \implies AB - A - B + I = BA - A - B + I \\ &\implies AB = BA\end{aligned}$$

Hence, $AB = BA$ if and only if $(A - I)(B - I) = (B - I)(A - I)$.

Solution 4.5.

1. Up. 2. Right.

3.

$$PJ + J^T P = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}.$$

It's not coincident. In Pascal Matrix, $a_{ij} = a_{i,j-1} + a_{i-1,j}$, where $i, j > 1$. PJ moves $a_{i,j-1}$ to a_{ij} , and $J^T P$ moves $a_{i-1,j}$ to a_{ij} , so the sum of PJ and $J^T P$ brings back the original a_{ij} again.

4.

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad J^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad J^3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad J^4 = O.$$

5.

Note that if $k > 3$ and $k \in \mathbb{N}$, then $J^k = O$. Thus,

$$(I - J)(I + J + J^2 + J^3) = I + J + J^2 + J^3 - J - J^2 - J^3 - J^4 = I.$$

6.

$$\begin{aligned}(J + I)^2 &= J^2 + 2J + I = \begin{bmatrix} 1 & 2 & 1 & 0 \\ & 1 & 2 & 1 \\ & & 1 & 2 \\ & & & 1 \end{bmatrix} \\ (J + I)^3 &= J^3 + 3J^2 + 3J + I = \begin{bmatrix} 1 & 3 & 3 & 1 \\ & 1 & 3 & 3 \\ & & 1 & 3 \\ & & & 1 \end{bmatrix} \\ (J + I)^4 &= J^4 + 4J^3 + 6J^2 + 4J + I = \begin{bmatrix} 1 & 4 & 6 & 4 \\ & 1 & 4 & 6 \\ & & 1 & 4 \\ & & & 1 \end{bmatrix} \\ (J + I)^k &= C_k^3 J^3 + C_k^2 J^2 + C_k^1 J + I = \begin{bmatrix} 1 & C_k^1 & C_k^2 & C_k^3 \\ & 1 & C_k^1 & C_k^2 \\ & & 1 & C_k^1 \\ & & & 1 \end{bmatrix}, k > 2, k \in \mathbb{N}.\end{aligned}$$

7.

If A commute with J , then A must be a 4×4 square matrix. Set $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$

By commutativity, we have

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So,

$$\begin{bmatrix} a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a_{11} & a_{12} & a_{13} \\ 0 & a_{21} & a_{22} & a_{23} \\ 0 & a_{31} & a_{32} & a_{33} \\ 0 & a_{41} & a_{42} & a_{43} \end{bmatrix}$$

It's obvious that $a_{21} = a_{31} = a_{41} = a_{42} = a_{43} = 0$, then

$$\begin{bmatrix} 0 & a_{22} & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a_{11} & a_{12} & a_{13} \\ 0 & 0 & a_{22} & a_{23} \\ 0 & 0 & a_{32} & a_{33} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence we have $\begin{cases} a_{32} = 0 \\ a_{11} = a_{22} = a_{33} = a_{44} \\ a_{12} = a_{23} = a_{34} \\ a_{24} = a_{13} \end{cases}$. Thus, $A = \begin{bmatrix} a & b & c & d \\ & a & b & c \\ & & a & b \\ & & & a \end{bmatrix}$, $a, b, c, d \in \mathbb{R}$.

Solution 4.6.

1. $\frac{1}{5} \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$.

2. $\begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 3 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$. The difference is rank one.

3. $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$. 4. $\begin{bmatrix} & & & 1 \\ & & \ddots & \\ & & & \\ 1 & & & \end{bmatrix}$.

5. Set $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} A & \\ & A \end{bmatrix}$. Thus,

$$\begin{bmatrix} A & \\ & A \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & \\ & A^{-1} \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 0 \\ \frac{3}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

6. $\frac{1}{4} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$. 7. $\begin{bmatrix} 1 & a & ab & abc \\ & 1 & b & bc \\ & & 1 & c \\ & & & 1 \end{bmatrix}$.

Solution 4.7.

1. $\forall A \in \mathbb{R}^{3 \times 3}, AB = 4A \implies$ each column of A is multiplied by 4 $\implies B = 4I_3$.

2. $\forall A \in \mathbb{R}^{3 \times 3}, BA = 4A \implies$ each row of A is multiplied by 4 $\implies B = 4I_3$.

3. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. 4. $\frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

5. $\begin{bmatrix} 0 & 1 \\ & 0 & 1 \\ & & 0 \end{bmatrix}$.

6.

Note that B must be a 2×2 square matrix. Set $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then,

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \begin{bmatrix} a+b & a+b \\ c+d & c+d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix} \\ &\implies \begin{cases} a+b = a+c \\ a+b = b+d \\ c+d = a+c \\ c+d = b+d \end{cases} \\ &\implies \begin{cases} b = c \\ a = d \end{cases} \\ &\implies B = \begin{bmatrix} m & n \\ n & m \end{bmatrix}, m, n \in \mathbb{R}. \end{aligned}$$